Signal Representation Using Bases

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Key Concepts

1) It is useful to describe signals as a weighted sum of basis or "building block" signals.

2) The discrete Fourier transform is an example where the basis signals are complex sinusoids of different frequencies.

3) The goal is to use basis functions that require a relatively small number of bases (and corresponding weights) to represent the signal or a class of signals that are of interest.

a) Compression is based on discarding basis components that are associated with small weights. This reduces the number of coefficients needed to (approximately) describe the signal.

b) Filtering is based on keeping weights (and bases) that contain "signal" while discarding weights (and bases) associated with noise.

c) Interference can be reduced by zeroing out bases that contain interference.

4) There is no universally optimal set of bases for all signals and all applications. Different bases are useful for different signals and problems.

5) There are three basic categories of basis expansions.

a) Fixed or deterministic bases are those that do not depend on the signal to be represented. Examples include sinusoids and wavelets.

b) Data adaptive bases use the statistics of the data to find bases. Examples include principle component analysis (PCA) and independent component analysis (ICA).

c) Parametric bases represent the signal as a function of an unknown parameter. For example, we may assume the signal is a sinusoid, but we don't know the frequency or phase of the sinusoid.

6) A signal with N values can be represented exactly by N linearly independent bases.

7) A set of bases is orthogonal if the inner product of any two bases in the set is zero.

8) If the bases are orthogonal, then the weight for each basis is obtained from the inner product of the signal with that basis.

Video Screenshots

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$x[n] = \sum_k a_k \psi_k[n]$ $\psi_k[n]$, $k=1,2,\ldots$ are basis signals (building blocks)

Example: Discrete Fourier Transform

$$\psi_k[n] = e^{jk\frac{2\pi}{N}n} \Rightarrow x[n] = \sum_{k=0}^{N-1} X[k]\, e^{jk\frac{2\pi}{N}n}$$

Goal: use basis that results in a low-dimesional (small number of nonzero a_k) representation of "signal" of interest

Compression: discard components associated with small a_k

Filtering: keep a_k that contain signal, discard a_k associated with noise

Interference reduction: zero out a_k containing interference

There is no universally optimal set of bases!

Categories

1) Fixed or deterministic.
   - Sinusoids with known frequencies (DFT)
   - Wavelets

2) Data adaptive. Use statistics of data to find good bases.
   - Principal Component Analysis (PCA)
   - Canonical Component Analysis (CCA)
   - Independent Component Analysis (ICA)

3) Parametric. Bases are known function of an unknown parameter.
   - Sinusoid with unknown frequency (ies)
   - Impulse with unknown onset time

Assume $\psi_k[n]$ are given.

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Define $\underline{x} = [x[0]\ x[1]\ \dots\ x[N-1]]^T$ $(N \times 1)$

$\underline{\psi}_k = [\psi_k[0]\ \psi_k[1]\ \dots\ \psi_k[N-1]]^T$, $k = 1, 2, \dots N$

$$\underline{x} = \sum_{k=1}^{N} a_k \underline{\psi}_k = \underbrace{[\underline{\psi}_1\ \underline{\psi}_2\ \dots\ \underline{\psi}_N]}_{\underline{\Psi}} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix} \Big\} \underline{a} = \underline{\Psi}\,\underline{a}$$

If $\underline{\psi}_k$ are linearly independent, N of them are sufficient to represent any $N \times 1$ signal.

$\Rightarrow \underline{\Psi}$ is invertible

$$\underline{a} = \underline{\Psi}^{-1}\underline{x} \longleftrightarrow \underline{x} = \underline{\Psi}\,\underline{a}$$

Important case: orthogonal bases

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$$\sum_n \psi_k[n]\,\psi_\ell^*[n] = 0 \quad \ell \neq k \Rightarrow \underline{\psi}_\ell^H \underline{\psi}_k = 0 \quad \ell \neq k$$

$$\underline{\underline{\Psi}}^{-1} = \begin{bmatrix} \underline{\psi}_1^H/(\underline{\psi}_1^H\underline{\psi}_1) \\ \underline{\psi}_2^H/(\underline{\psi}_2^H\underline{\psi}_2) \\ \vdots \\ \underline{\psi}_N^H/(\underline{\psi}_N^H\underline{\psi}_N) \end{bmatrix}, \text{ so } \underline{a} = \underline{\underline{\Psi}}^{-1}\underline{x} \Rightarrow a_k = \frac{1}{\underline{\psi}_k^H\underline{\psi}_k}\underline{\psi}_k^H\underline{x}$$

$$= \frac{1}{\underline{\psi}_k^H\underline{\psi}_k}\sum_{n=0}^{N-1}\psi_k^*[n]x[n]$$

Example: DFT

$$\psi_k[n] = e^{jk\frac{2\pi}{N}n}$$

$$\underline{\psi}_\ell^H\underline{\psi}_k = \sum_{n=0}^{N-1} e^{-j\ell\frac{2\pi}{N}n}\, e^{jk\frac{2\pi}{N}n} = \begin{cases} 0 & \ell \neq k \\ N & \ell = k \end{cases} \quad (\underline{\psi}_\ell^H\underline{\psi}_\ell)$$

$$a_k = \frac{1}{\underline{\psi}_k^H\underline{\psi}_k}\sum_{n=0}^{N-1}\psi_k^*[n]x[n] \Rightarrow a_k = \frac{1}{N}\sum_{n=0}^{N-1} e^{-jk\frac{2\pi}{N}n}x[n]$$

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